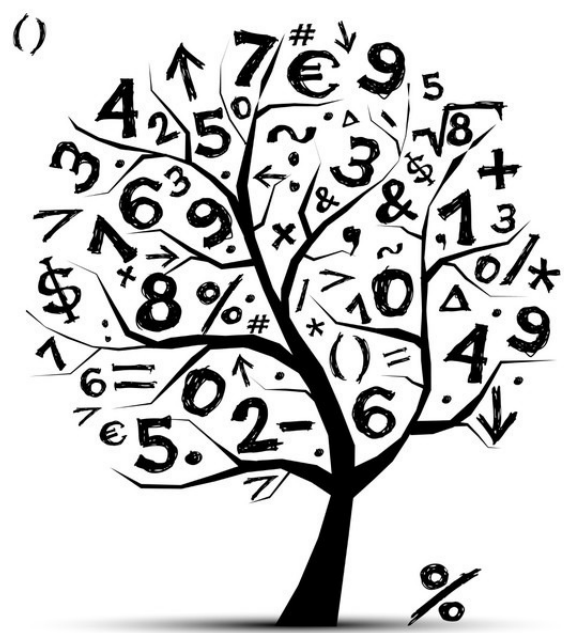


Last update: October 7, 2026

Matrices



Contents

1	Definition and transpose	2
1.1	Rows and columns	3
1.2	Minor matrices	3
2	Sum of matrices and scalar multiplication	4
2.1	Properties	5
3	Matrix product	6
3.1	Properties	8
4	Special matrices	9
4.1	Permutation matrices	12
4.2	Inverse matrices	13
4.3	Semidefinite matrices	14

5	Determinant	15
5.1	Laplace expansion	16
5.2	Determinant of triangular and diagonal matrices	16
5.3	Properties of determinants	18
6	Rank	20

1 Definition and transpose

Definition 1: Matrix

A **matrix** is a rectangular table of real numbers with m rows and n columns.

Given a matrix $A \in \mathbb{R}^{m \times n}$, we write:

$$A = \begin{pmatrix} [A]_{1,1} & [A]_{1,2} & \cdots & [A]_{1,n} \\ [A]_{2,1} & [A]_{2,2} & \cdots & [A]_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ [A]_{m,1} & [A]_{m,2} & \cdots & [A]_{m,n} \end{pmatrix} \quad \text{or, for brevity,} \quad A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

- The integers m and n are the **dimensions** of the matrix: m is the number of rows and n is the number of columns.
- The quantity $[A]_{i,j}$ (or a_{ij}) denotes the **entry** of A in row i and column j , for $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$.

Example 1: Matrix

The matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

is a matrix with $m = 2$ rows and $n = 3$ columns, where $a_{11} = 1$, $a_{12} = 2$, $a_{13} = 3$, $a_{21} = 4$, $a_{22} = 5$, and $a_{23} = 6$.

Definition 2: Transpose of a matrix

Given a matrix $A \in \mathbb{R}^{m \times n}$, the **transpose** of A , denoted by A' , is the $n \times m$ matrix obtained by interchanging rows and columns:

$$A' = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{n \times m}. \quad (1)$$

- The entry in position (i, j) of the transpose matrix A' is:

$$a'_{ij} = a_{ji}, \quad \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, m\}.$$

Example 2: Transpose of a matrix

Given the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3},$$

its transpose is

$$A' = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

1.1 Rows and columns

- The i -th row of a matrix A is denoted by

$$R_i(A) \quad \text{or} \quad A_i$$

- The j -th column of a matrix A is denoted by

$$C_j(A) \quad \text{or} \quad A_j$$

- In both cases, we can simply use R_i or C_j when the matrix is clear from the context.

1.2 Minor matrices

- Let $A \in \mathbb{R}^{n \times m}$ be a matrix and let $i \in \{1, 2, \dots, n\}$, $j \in \{1, 2, \dots, m\}$. We denote by A_{ij} the $(n-1) \times (m-1)$ matrix obtained from A by deleting row i and column j . The matrix A_{ij} is called the **minor matrix** associated with the entry a_{ij} .

Example 3: Minor matrix

Let

$$A = \begin{pmatrix} 1 & -2 & 0 & 5 \\ 3 & 4 & 7 & -1 \\ 2 & 0 & -3 & 6 \\ -4 & 1 & 2 & 8 \end{pmatrix} \in \mathbb{R}^{4 \times 4}.$$

For $i = 2$ and $j = 3$, the minor matrix $A_{23} \in \mathbb{R}^{3 \times 3}$ is obtained by deleting row 2 and column 3:

$$A_{23} = \begin{pmatrix} 1 & -2 & 5 \\ 2 & 0 & 6 \\ -4 & 1 & 8 \end{pmatrix}.$$

2 Sum of matrices and scalar multiplication

Given two matrices

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_A \in \mathbb{R}^{m \times n} \quad \text{and} \quad \underbrace{\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}}_B \in \mathbb{R}^{m \times n}$$

their sum is the matrix

$$A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix} \quad (2)$$

Given a scalar $\lambda \in \mathbb{R}$ and a matrix

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_A \in \mathbb{R}^{m \times n}$$

the **scalar multiplication** of A by λ is the matrix

$$\lambda A = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{pmatrix} \quad (3)$$

Example 4: Sum of matrices and scalar multiplication

Consider the matrices

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 5 \\ -1 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then

$$A + B = \begin{pmatrix} 1+4 & -2+5 \\ 3+(-1) & 0+2 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 2 & 2 \end{pmatrix},$$

and, for $\lambda = -2$,

$$\lambda A = -2 \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ -6 & 0 \end{pmatrix}.$$

2.1 Properties

The main properties of matrix sum and scalar multiplication are:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (4)$$

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}), \quad \forall \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{m \times n} \quad (5)$$

$$\mathbf{A} + \mathbf{0} = \mathbf{A}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (6)$$

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (7)$$

$$\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}, \quad \forall \lambda \in \mathbb{R}, \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (8)$$

$$(\lambda + \mu)\mathbf{A} = \lambda\mathbf{A} + \mu\mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (9)$$

$$\lambda(\mu\mathbf{A}) = (\lambda\mu)\mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (10)$$

$$1 \cdot \mathbf{A} = \mathbf{A} \quad \text{and} \quad 0 \cdot \mathbf{A} = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n}. \quad (11)$$

3 Matrix product

Given two matrices

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_A \in \mathbb{R}^{m \times n} \quad \text{and} \quad \underbrace{\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix}}_B \in \mathbb{R}^{n \times p}$$

their multiplication is the matrix

$$AB = \begin{pmatrix} \sum_{j=1}^n a_{1j}b_{j1} & \sum_{j=1}^n a_{1j}b_{j2} & \cdots & \sum_{j=1}^n a_{1j}b_{jp} \\ \sum_{j=1}^n a_{2j}b_{j1} & \sum_{j=1}^n a_{2j}b_{j2} & \cdots & \sum_{j=1}^n a_{2j}b_{jp} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n a_{mj}b_{j1} & \sum_{j=1}^n a_{mj}b_{j2} & \cdots & \sum_{j=1}^n a_{mj}b_{jp} \end{pmatrix} \in \mathbb{R}^{m \times p} \quad (12)$$

where the entry in position (i, k) , i.e., $[AB]_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$, is the scalar product of the i -th row of A and the k -th column of B .

Example 5: Matrix product

Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{and} \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

The product AB is:

$$\begin{aligned} AB &= \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} \\ &= \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \end{aligned}$$

The product BA is:

$$\begin{aligned} BA &= \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix} \end{aligned}$$

Example 6: Matrix product

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 3} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 2}$$

The product $\mathbf{AB}$ is:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 3 & 1 \cdot 1 + 2 \cdot 0 + (-1) \cdot 5 \\ 0 \cdot 2 + 3 \cdot (-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 5 \end{pmatrix} \\ &= \begin{pmatrix} -3 & -4 \\ 9 & 20 \end{pmatrix} \end{aligned}$$

The product $\mathbf{BA}$ is:

$$\begin{aligned} \mathbf{BA} &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot 3 & 2 \cdot (-1) + 1 \cdot 4 \\ -1 \cdot 1 + 0 \cdot 0 & -1 \cdot 2 + 0 \cdot 3 & -1 \cdot (-1) + 0 \cdot 4 \\ 3 \cdot 1 + 5 \cdot 0 & 3 \cdot 2 + 5 \cdot 3 & 3 \cdot (-1) + 5 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 7 & 2 \\ -1 & -2 & 1 \\ 3 & 21 & 17 \end{pmatrix} \end{aligned}$$

In particular, $\mathbf{AB} \neq \mathbf{BA}$.

3.1 Properties

The main properties of the matrix product (when dimensions are compatible) are:

$$(AB)C = A(BC), \quad \forall A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p}, C \in \mathbb{R}^{p \times q} \quad (13)$$

$$A(B + C) = AB + AC, \quad \forall A \in \mathbb{R}^{m \times n}, B, C \in \mathbb{R}^{n \times p} \quad (14)$$

$$(A + B)C = AC + BC, \quad \forall A, B \in \mathbb{R}^{m \times n}, C \in \mathbb{R}^{n \times p} \quad (15)$$

$$\lambda(AB) = (\lambda A)B = A(\lambda B), \quad \forall \lambda \in \mathbb{R}, \forall A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p} \quad (16)$$

$$I_m A = A \quad \text{and} \quad A I_n = A, \quad \forall A \in \mathbb{R}^{m \times n} \quad (17)$$

In general, the matrix product is **not commutative**:

$$AB \neq BA \quad \text{in general.} \quad (18)$$

4 Special matrices

Definition 3: Square matrix

A matrix $A \in \mathbb{R}^{m \times n}$ is called a **square matrix** if it has the same number of rows and columns ($m = n$). In this case, $A \in \mathbb{R}^{n \times n}$ and it has the form:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}. \quad (19)$$

Example 7: Square matrix

An example of a square matrix of size 3 is:

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \\ 1 & 0 & 7 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Definition 4: Identity matrix

A square matrix $I_n \in \mathbb{R}^{n \times n}$ is called the **identity matrix** if it has ones on the main diagonal and zeros elsewhere. It has the form:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \quad (20)$$

- Equivalently, the identity matrix satisfies

$$[I_n]_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases} \quad \forall i, j \in \{1, 2, \dots, n\}.$$

Example 8: Identity matrix

The identity matrix of size 3 is:

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

- For any matrix $A \in \mathbb{R}^{m \times n}$, we have:

$$I_m A = A \quad \text{and} \quad A I_n = A.$$

Definition 5: Diagonal matrix

A square matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ is called **diagonal** if all its off-diagonal entries are zero. It has the form:

$$\mathbf{D} = \begin{pmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{pmatrix}. \quad (21)$$

- Equivalently, a diagonal matrix satisfies

$$d_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i \neq j.$$

Example 9: Diagonal matrix

The matrix

$$\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is diagonal.

Definition 6: Symmetric matrix

A square matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is called **symmetric** if it has the same entries with respect to the main diagonal (i.e., the entry in position (i, j) equals the entry in position (j, i)). It has the form:

$$\mathbf{Q} = \begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{12} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{1n} & q_{2n} & \cdots & q_{nn} \end{pmatrix}. \quad (22)$$

- Equivalently, a symmetric matrix satisfies

$$q_{ij} = q_{ji}, \quad \forall i, j \in \{1, 2, \dots, n\}.$$

- In matrix form, symmetry can be written as

$$\mathbf{Q}' = \mathbf{Q}.$$

Definition 7: Upper triangular matrix

A square matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ is called **upper triangular** if all its entries below the main diagonal are zero. It has the form:

$$\mathbf{U} = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}. \quad (23)$$

- Equivalently, an upper triangular matrix satisfies

$$u_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i > j.$$

Example 10: Upper triangular matrix

The matrix

$$\mathbf{U} = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & -2 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is upper triangular.

Definition 8: Lower triangular matrix

A square matrix $\mathbf{L} \in \mathbb{R}^{n \times n}$ is called **lower triangular** if all its entries above the main diagonal are zero. It has the form:

$$\mathbf{L} = \begin{pmatrix} \ell_{11} & 0 & \cdots & 0 \\ \ell_{21} & \ell_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{nn} \end{pmatrix}. \quad (24)$$

- Equivalently, a lower triangular matrix satisfies

$$\ell_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i < j.$$

Example 11: Lower triangular matrix

The matrix

$$\mathbf{L} = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is lower triangular.

4.1 Permutation matrices

- Permutation matrices are square matrices obtained by permuting the rows (or equivalently, the columns) of the identity matrix. They encode permutations of indices and implement row/column reorderings through matrix multiplication.

Given an ordering (permutation) π of $\{1, \dots, n\}$, we denote by $\mathbf{P}_\pi$ the **permutation matrix** obtained from $\mathbf{I}_n$ by reordering its rows according to π , i.e., placing row $\pi(1)$ first, then row $\pi(2)$, ..., and finally row $\pi(n)$.

- **Row reordering by left multiplication.** Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{m \times m}$, the matrix

$$\mathbf{P}_\pi \mathbf{A}$$

consists of the rows of $\mathbf{A}$ reordered according to the permutation π (row $\pi(1)$ becomes the first row, row $\pi(2)$ becomes the second row, etc.).

- **Column reordering by right multiplication.** Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{n \times n}$, the matrix

$$\mathbf{A} \mathbf{P}'_\pi$$

(note the transpose: $\mathbf{P}'_\pi$ is obtained from $\mathbf{I}_n$ by reordering its *columns* according to π) consists of the columns of $\mathbf{A}$ reordered according to the permutation π (column $\pi(1)$ becomes the first column, column $\pi(2)$ becomes the second column, etc.).

Example 12: Row reordering via permutation matrix

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

Given the desired row ordering $\pi = (3, 1, 2)$ (i.e., row 3 first, then row 1, then row 2), we use the permutation matrix $\mathbf{P}_\pi$ obtained from $\mathbf{I}_3$ by reordering its rows as $(3, 1, 2)$:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then:

$$\mathbf{P}_\pi \mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The rows of $\mathbf{A}$ have been reordered according to $\pi = (3, 1, 2)$.

Example 13: Column reordering via permutation matrix

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

Given the desired column ordering $\pi = (2, 3, 1)$ (i.e., column 2 first, then column 3, then column 1), we use the permutation matrix P_π obtained from I_3 by reordering its rows as $(2, 3, 1)$, and its transpose:

$$P_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad P'_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then:

$$AP'_\pi = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{pmatrix}.$$

The columns of A have been reordered according to $\pi = (2, 3, 1)$.

4.2 Inverse matrices**Definition 9: inverse matrix**

A square matrix $A \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that

$$AA^{-1} = I \quad \text{and} \quad A^{-1}A = I.$$

Such a matrix A^{-1} is called the **inverse** of A .

- If it exists, the inverse matrix A^{-1} is unique.

4.3 Semidefinite matrices

Definition 10: semidefinite matrices

A square **symmetric** matrix $Q \in \mathbb{R}^{n \times n}$ is

- **positive semidefinite** if

$$\mathbf{x}' Q \mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- **positive definite** if

$$\mathbf{x}' Q \mathbf{x} > 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$$

- **negative semidefinite** if

$$\mathbf{x}' Q \mathbf{x} \leq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- **negative definite** if

$$\mathbf{x}' Q \mathbf{x} < 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$$

Example 14: positive semidefinite matrix

- The identity matrix

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + x_2^2 \geq 0, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

- The matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{aligned} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 \\ &= x_1^2 + (x_1 - x_2)^2 + x_2^2 \geq 0, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2. \end{aligned}$$

5 Determinant

- **Determinant of a 1×1 matrix.** Let

$$A = (a) \in \mathbb{R}^{1 \times 1}.$$

Then

$$\det(A) = a. \quad (25)$$

- **Determinant of a 2×2 matrix.** Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then

$$\det(A) = ad - bc. \quad (26)$$

- **Determinant of a 3×3 matrix (Sarrus rule).** Let

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then

$$\det(A) = aei + bfg + cdh - ceg - bdi - afh. \quad (27)$$

Example 15: determinant of a 2×2 matrix

Let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

Then

$$\det(A) = 2 \cdot 4 - (-1) \cdot 3 = 8 + 3 = 11.$$

Example 16: determinant of a 3×3 matrix

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}.$$

Using Sarrus:

$$\det(A) = 1 \cdot 1 \cdot 1 + 2 \cdot 4 \cdot 2 + 0 \cdot 3 \cdot (-1) - 0 \cdot 1 \cdot 2 - 2 \cdot 3 \cdot 1 - 1 \cdot 4 \cdot (-1).$$

Hence

$$\det(A) = 1 + 16 + 0 - 0 - 6 + 4 = 15.$$

5.1 Laplace expansion

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. The determinant $\det(A)$ can be computed recursively by **Laplace expansion** along any row or column.

For any row $i \in \{1, 2, \dots, n\}$, we have:

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij} \quad (28)$$

For any column $j \in \{1, 2, \dots, n\}$, we have:

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij} \quad (29)$$

Where C_{ij} is the **cofactor** of the entry a_{ij} , defined by

$$C_{ij} = (-1)^{i+j} \det(A_{ij}) \quad (30)$$

where A_{ij} is the minor matrix obtained from A by deleting row i and column j .

Example 17: Determinant via Laplace expansion

Let

$$A = \begin{pmatrix} -2 & 2 & -3 \\ -1 & 1 & 3 \\ 2 & 0 & -1 \end{pmatrix}.$$

We expand along the second column (it contains a zero):

$$\det(A) = 2 C_{12} + 1 C_{22} + 0 \cdot C_{32}.$$

Compute the cofactors:

$$C_{12} = (-1)^{1+2} \det \begin{pmatrix} -1 & 3 \\ 2 & -1 \end{pmatrix} = -((-1)(-1) - 2 \cdot 3) = -(1 - 6) = 5,$$

$$C_{22} = (-1)^{2+2} \det \begin{pmatrix} -2 & -3 \\ 2 & -1 \end{pmatrix} = ((-2)(-1) - 2(-3)) = 2 + 6 = 8.$$

Therefore,

$$\det(A) = 2 \cdot 5 + 1 \cdot 8 = 18.$$

5.2 Determinant of triangular and diagonal matrices

- For triangular matrices (upper or lower), the determinant has a particularly simple form: it is the product of the diagonal entries.

If $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**, then

$$\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}. \quad (31)$$

If $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix**, then

$$\det(\mathbf{L}) = \prod_{i=1}^n \ell_{ii}. \quad (32)$$

- In particular, for a **diagonal matrix** $\mathbf{D} \in \mathbb{R}^{n \times n}$:

$$\det(\mathbf{D}) = \prod_{i=1}^n d_{ii}.$$

- For the **identity matrix** $\mathbf{I}_n$:

$$\det(\mathbf{I}_n) = 1.$$

Example 18: Determinant of an upper triangular matrix

Let

$$\mathbf{U} = \begin{pmatrix} 2 & 3 & -1 \\ 0 & -4 & 5 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{U}) = 2 \cdot (-4) \cdot 3 = -24.$$

Example 19: Determinant of a lower triangular matrix

Let

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 4 & -2 \end{pmatrix}.$$

Then

$$\det(\mathbf{L}) = 1 \cdot 3 \cdot (-2) = -6.$$

Example 20: Determinant of a diagonal matrix

Let

$$\mathbf{D} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{D}) = 5 \cdot (-2) \cdot 3 = -30.$$

5.3 Properties of determinants

- The determinant satisfies several important algebraic properties that make its computation and application more efficient.

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ be square matrices and $\lambda \in \mathbb{R}$ be a scalar. Then:

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B}) \quad (33)$$

$$\det(\mathbf{A}') = \det(\mathbf{A}) \quad (34)$$

$$\det(\lambda \mathbf{A}) = \lambda^n \det(\mathbf{A}) \quad (35)$$

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}, \quad \text{if } \mathbf{A} \text{ is invertible} \quad (36)$$

$$\det(\mathbf{I}_n) = 1 \quad (37)$$

- From properties (33) and (37), if $\mathbf{A}$ is invertible:

$$\det(\mathbf{A}) \cdot \det(\mathbf{A}^{-1}) = \det(\mathbf{AA}^{-1}) = \det(\mathbf{I}_n) = 1,$$

which gives property (36).

- Property (35) shows that scalar multiplication affects the determinant by the n -th power of the scalar (not linearly).
- Property (34) implies that row operations and column operations have symmetric effects on the determinant.

Example 21: Determinant of a product

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 0 = 6, \quad \det(\mathbf{B}) = 1 \cdot 2 - (-1) \cdot 4 = 6.$$

The product is:

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 12 & 6 \end{pmatrix}.$$

Then:

$$\det(\mathbf{AB}) = 6 \cdot 6 - 0 \cdot 12 = 36 = \det(\mathbf{A}) \cdot \det(\mathbf{B}).$$

Example 22: Determinant of a transpose

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{A}' = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\mathbf{A}') = 1 \cdot 4 - 3 \cdot 2 = -2.$$

Therefore, $\det(\mathbf{A}') = \det(\mathbf{A})$.

Example 23: Determinant of a scalar multiple

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad \lambda = 2.$$

Then:

$$\lambda \mathbf{A} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\lambda \mathbf{A}) = 2 \cdot 8 - 4 \cdot 6 = -8.$$

Since $n = 2$, we verify:

$$\det(\lambda \mathbf{A}) = -8 = 2^2 \cdot (-2) = \lambda^2 \det(\mathbf{A}).$$

Example 24: Determinant of an inverse

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 1 = 1.$$

The inverse matrix is:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Then:

$$\det(\mathbf{A}^{-1}) = 1 \cdot 2 - (-1) \cdot (-1) = 1 = \frac{1}{\det(\mathbf{A})}.$$

A matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

6 Rank

Definition 11: rank of a matrix

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The **rank** of $\mathbf{A}$, denoted by $\text{rank}(\mathbf{A})$, is the maximum number of linearly independent columns of $\mathbf{A}$.

- The columns of $\mathbf{A}$ are vectors of $\mathbb{R}^m$, and linear independence is meant in the sense of the definition given in the chapter on vectors: $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$ implies $\lambda_1 = \dots = \lambda_k = 0$.
- Equivalently, $\text{rank}(\mathbf{A})$ is the maximum number of linearly independent rows of $\mathbf{A}$.
- We always have

$$\text{rank}(\mathbf{A}) \leq \min\{m, n\}.$$

- If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, then

$$\text{rank}(\mathbf{A}) = n \iff \det(\mathbf{A}) \neq 0.$$

Example 25: rank of a matrix

Let

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

The second row is a multiple of the first one:

$$(2 \quad 4 \quad 6) = 2(1 \quad 2 \quad 3).$$

Hence the two rows are linearly dependent, so there is only one linearly independent row. Therefore,

$$\text{rank}(\mathbf{B}) = 1.$$

Example 26: rank of a 3×3 matrix

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

The third column is the sum of the first two:

$$C_3(\mathbf{A}) = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = C_1(\mathbf{A}) + C_2(\mathbf{A}),$$

hence the three columns are linearly dependent and $\text{rank}(\mathbf{A}) \leq 2$. On the other hand, the first two columns are linearly independent: $\lambda_1 C_1(\mathbf{A}) + \lambda_2 C_2(\mathbf{A}) = \mathbf{0}$ gives $\lambda_1 = 0$ (first entry) and $\lambda_2 = 0$ (third entry). Therefore,

$$\text{rank}(\mathbf{A}) = 2.$$

Consistently, expanding along the first row, $\det(\mathbf{A}) = 1 \cdot (1 \cdot 1 - 3 \cdot 1) - 0 + 1 \cdot (2 \cdot 1 - 1 \cdot 0) = -2 + 2 = 0$.